

# **Dynamic Programming and Applications**

Heterogeneous Agents and Inequality

Lecture 11

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# Outline

Part 2: Huggett (1993) in continuous time (Bewley-Huggett-Aiyagari)

*This is the textbook heterogeneous agent model / standard incomplete markets model*

1. Households, firms and market clearing
2. Competitive equilibrium in sequence form
3. Recursive representation using dynamic programming
4. Competitive equilibrium in recursive form
5. Stationary competitive equilibrium in recursive form

I learned all this from Benjamin Moll!

*Income and Wealth Distribution in Macroeconomics: A Continuous-Time Approach*

See also Ben's teaching material: <https://benjaminmoll.com/lectures/>

## Part 2: Huggett (1993)

# Model overview

- Time is continuous,  $t \in [0, \infty)$
- No aggregate uncertainty  $\implies$  focus on one-time, unanticipated (“MIT”) shocks (perfect foresight wrt. macroeconomic aggregates)
- Two types of agents: continuum of households (measure 1) + representative firm
- Households face “uninsurable idiosyncratic income risk”
- There is a single riskfree asset in zero net supply ( $\sim$  government bond)

## Plan:

1. Present model in sequence form focusing exposition on individual  $i$
2. Recursive representation of competitive equilibrium

# Households

**Preferences.** The individual lifetime utility of a household  $i \in [0, 1]$  is

$$V_{i,0} = \max_{\{c_{i,t}\}_{t \geq 0}} \mathbb{E}_0 \int_0^\infty e^{-\rho t} u(c_{i,t}) dt$$

**Budget and borrowing constraints:**

$$\dot{a}_{i,t} = w_t y_{i,t} + r_t a_{i,t} - c_{i,t}$$

$$a_{i,t} \geq \underline{a}$$

**Idiosyncratic income risk:** each  $y_{i,t}$  follows a Markov chain (later: diffusion)

$$y_{i,t} \in \{y_1, y_2\} \text{ Poisson with intensities } \lambda_1, \lambda_2$$

**Definition.** The problem of household  $i$  (in sequence form) is

$$\max_{\{c_{i,t}\}_{t \geq 0}} \mathbb{E}_0 \int_0^\infty e^{-\rho t} u(c_{i,t}) dt \quad \text{s.t.}$$

$$\dot{a}_{i,t} = w_t y_{i,t} + r_t a_{i,t} - c_{i,t}$$

$$y_{i,t} \in \{y_1, y_2\} \text{ Poisson with intensities } \lambda_1, \lambda_2$$

$$a_{i,t} \geq \underline{a}$$

taking as given initial  $(a_{i,0}, y_{i,0})$

A **solution** to the household problem is a stochastic process  $\{c_{i,t}, a_{i,t}\}_{t \geq 0}$

# Firms

- A representative firm produces the (homogeneous) final consumption good using technology

$$Y_t = A_t \ell_t$$

- Firms are small and perfectly competitive  $\implies$  firm problem is

$$\max_{\ell_t} Y_t - w_t \ell_t$$

- Firm problem is **static**  $\implies$  otherwise we would have to think hard about ownership

# Markets

How many markets are there?



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**Goods market:**

$$Y_t = \int_0^1 c_{i,t} di$$

**Labor market:**

$$\ell_t = \int_0^1 y_{i,t} di$$

**Asset market:**

$$0 = \int_0^1 a_{i,t} di$$

# Competitive Equilibrium

**Definition.** (Competitive equilibrium: sequence form) *Taking as given an initial distribution of assets and individual labor productivities  $\{a_{i,0}, y_{i,0}\}_i$  as well as an exogenous path for TFP  $\{A_t\}$ , a competitive equilibrium comprises an allocation  $\{Y_t, \ell_t, c_{i,t}, a_{i,t}\}$  and prices  $\{r_t, w_t\}$  such that: (i) households optimize, (ii) firms optimize, and (iii) markets clear.*

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- Always show definition of your equilibrium!
- Why is this a definition of competitive equilibrium “in sequence form”?
- How many parts are there to this definition?

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- Right now: model = stochastic processes for each  $i$   
Recursive representation = functions over state variables
- What about the household problem in PE? What are the state variables? Firms?  
*What's the difference between PE and GE?*

# Recursive representation

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- Let's think hard: what does it take to bring this GE model into recursive form?
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Recursive representation = functions over state variables
- What about the household problem in PE? What are the state variables? Firms?  
*What's the difference between PE and GE?*
- What about markets and GE? What does  $\int_0^1 c_{i,t} di$  mean?

Recall household problem: Given  $(a_{i,0}, y_{i,0})$ ,

$$\max_{\{c_{i,t}\}_{t \geq 0}} \mathbb{E}_0 \int_0^\infty e^{-\rho t} u(c_{i,t}) dt \quad \text{s.t.}$$

$$\dot{a}_{i,t} = w_t y_{i,t} + r_t a_{i,t} - c_{i,t}$$

$$y_{i,t} \in \{y_1, y_2\} \text{ Poisson with intensities } \lambda_1, \lambda_2$$

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**Recursive representation:**

$$\rho V_t(a, y) = \max_c \left\{ u(c) + \mathbb{E}_t \frac{dV_t(a, y)}{dt} \right\}$$

Two **Qs**: What is  $\mathbb{E}_t \frac{dV_t(a, y)}{dt}$ ? And what about borrowing constraint?

From previous slide:

$$\rho V_t(a, y) = \max_c \left\{ u(c) + \mathbb{E}_t \frac{dV_t(a, y)}{dt} \right\}$$

**Q1:** What is continuation value?

From previous slide:

$$\rho V_t(a, y) = \max_c \left\{ u(c) + \mathbb{E}_t \frac{dV_t(a, y)}{dt} \right\}$$

**Q1:** What is continuation value?

$$\rho V_t(a, y_j) = \max_c \left\{ u(c) + (r_t a + w_t y_j - c) \partial_a V_t(a, y_j) + \lambda_j (V_t(a, y_{-j}) - V_t(a, y_j)) + \partial_t V_t(a, y_j) \right\}$$

Resolving max operator gives FOC, which defines **consumption policy function**:

$$u'(c_t(a, y_j)) = \partial_a V_t(a, y_j)$$

for all  $a$  and  $j$ . Define **savings policy function** as  $s_t(a, y_j) = r_t a + w_t y_j - c_t(a, y_j)$

**Q2:** Where is the borrowing constraint  $a_{i,t} \geq \underline{a}$  in the HJB?

**Answer:** in the boundary condition!

- Borrowing constraint gives rise to **state constraint boundary condition**

$$\partial_a V_t(\underline{a}, y_j) \geq u'(r_t \underline{a} + w_t y_j)$$

- Economic intuition: value of saving must be weakly larger than value of consuming
- Heuristic derivation: the FOC still holds at the borrowing constraint

$$u'(c_t(\underline{a}, j)) = \partial_a V_t(a, y_j)$$

- But borrowing constraint requires that

$$s_t(\underline{a}, y_j) = r_t \underline{a} + w_t y_j - c_t(\underline{a}, y_j) \geq 0$$

- Borrowing constraint showing up as boundary condition = major advantage of continuous time!

**Summary:** A solution to the household problem in **recursive form** is a set of two functions  $V_t(a, y)$  and  $c_t(a, y)$  that satisfy

$$\rho V_t(a, y_j) = u(c_t(a, y_j)) + (r_t a + w_t y_j - c_t(a, y_j)) \partial_a V_t(a, y_j) + \lambda_j (V_t(a, y_{-j}) - V_t(a, y_j)) + \partial_t V_t(a, y_j)$$

$$u'(c_t(a, y_j)) = \partial_a V_t(a, y_j)$$

with HJB boundary condition

$$\partial_a V_t(\underline{a}, y_j) \geq u'(r_t \underline{a} + w_t y_j)$$

To save space, will now use savings policy function  $s_t(a, y_j) \equiv r_t a + w_t y_j - c_t(a, y_j)$  as shorthand

# Income and Wealth Distribution

- We now have recursive representations of the household (and firm) problem
- But how do we do GE? We have conditions like  $Y_t = \int_0^1 c_{i,t} di$
- Instead, we want to get **aggregate consumption** by integrating over  $c_t(a, y)$
- Issue: there may be many households  $i$  in state  $(a, y)$ ! Remember (important): household  $i$  is **uniquely** identified by her states  $(a_i, y_i)$  in this model
- Solution: integrate against the joint density  $\sim$  income-wealth distribution  $g_t(a, y)$ :

$$Y_t = \iint c_t(a, y) g_t(a, y) da dy \equiv \sum_j \int c_t(a, y_j) g_t(a, y_j) da$$

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 $\implies$  Need new equilibrium condition!
- Where does this equilibrium condition come from?



# Kolmogorov Forward Equation

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- Notice:  $g_t(a, y)$  is a new object that wasn't part of the eq. definition in sequence form  
 $\implies$  Need new equilibrium condition!
- Where does this equilibrium condition come from?  
 $\implies g_t(a, y)$  must be **consistent with household behavior**
- Turns out:  $g_t(a, y)$  solves a **Kolmogorov forward equation**
- Another major advantage of continuous time!

**Result:** the joint density  $g_t(a, y)$  solves the Kolmogorov forward (KF) equation

$$\partial_t g_t(a, y_j) = -\partial_a \left[ (r_t a + w_t y_j - c_t(a, y_j)) g_t(a, y_j) \right] - \lambda_j g_t(a, y_j) + \lambda_{-j} g_t(a, y_{-j})$$

**Proof:** Define aggregate consumption as cross-sectional average consumption

$$C_t = \mathbb{E}_i(c_{i,t}) = \mathbb{E}_g(c_t(a, y)) \equiv \iint c_t(a, y) g_t(a, y) da dy$$

By “Ito’s lemma”, we have

$$dc_t(a, y_j) = \partial_t c_t(a, y_j) dt + s_t(a, y_j) \partial_a c_t(a, y_j) dt + \lambda_j (c_t(a, y_{-j}) - c_t(a, y_j)) dt$$

On the **one hand**, we have

$$dC_t = \mathbb{E}_g(dc_t(a, y)) = \iint dc_t(a, y)g_t(a, y) da dy$$

and so plugging in

$$= \sum_j \int \left[ \partial_t c_t(a, y_j) + s_t(a, y_j) \partial_a c_t(a, y_j) + \lambda_j (c_t(a, y_{-j}) - c_t(a, y_j)) \right] g_t(a, y_j) da dt$$

On the **other hand**, we have

$$dC_t = d \iint c_t(a, y)g_t(a, y) da dy = \sum_j \int \left[ g_t(a, y_j) \partial_t c_t(a, y_j) + c_t(a, y_j) \partial_t g_t(a, y_j) \right] dadt$$

We now **equate the two** (next slide)

$$0 = \sum_j \int \left\{ \left[ s_t(a, y_j) \partial_a c_t(a, y_j) + \lambda_j (c_t(a, y_{-j}) - c_t(a, y_j)) \right] g_t(a, y_j) - \left[ c_t(a, y_j) \partial_t g_t(a, y_j) \right] \right\} da dt$$

Integrating the first term by parts:

$$\sum_j \int \left[ -\partial_a [s_t(a, y_j) g_t(a, y_j)] - \lambda_j g_t(a, y_j) + \lambda_{-j} g_t(a, y_{-j}) \right] c_t(a, y) da dt$$

(plus boundary conditions: abstract from those for now)

Finally arrive at:

$$0 = \sum_j \int \left[ -\partial_t g_t(a, y_j) - \partial_a [s_t(a, y_j) g_t(a, y_j)] - \lambda_j g_t(a, y_j) + \lambda_{-j} g_t(a, y_{-j}) \right] c_t(a, y_j) da dt$$

Concluding: this must hold “for all”  $c_t(a, y)$ , so term in brackets = 0

# Competitive Equilibrium: Recursive Form

**Definition.** Taking as given an initial joint density  $g_0(a, y)$  and an exogenous path of TFP  $\{A_t\}$ , a competitive equilibrium (in recursive form) comprises **functions**

$$\left\{ V_t(a, y), c_t(a, y), g_t(a, y) \right\} \quad \text{and} \quad \left\{ Y_t, \ell_t, r_t, w_t \right\}$$

such that (i) households optimize, (ii) firms optimize, (iii) markets clear, and (iv) the joint density evolves consistently with household behavior.

**HJB and FOC:**

$$\rho V_t(a, y_j) = u(c_t(a, y_j)) + s_t(a, y_j) \partial_a V_t(a, y_j) + \lambda_j (V_t(a, y_{-j}) - V_t(a, y_j)) + \partial_t V_t(a, y_j)$$

$$\partial_a V_t(\underline{a}, y_j) \geq u'(r_t \underline{a} + A_t y_j)$$

$$u'(c_t(a, y_j)) = \partial_a V_t(a, y_j)$$

**KF:** 
$$\partial_t g_t(a, y_j) = -\partial_a \left[ s_t(a, y_j) g_t(a, y_j) \right] - \lambda_j g_t(a, y_j) + \lambda_{-j} g_t(a, y_{-j})$$

**Bond market:**

$$0 = \sum_j \int a g_t(a, y_j) da$$

(We plugged in for  $w_t = A_t$  and dropped goods market clearing by Walras' law)

In **continuous time**: HA models = system of 2 coupled PDEs!

# Stationary Competitive Equilibrium

**Definition.** With  $A_t = A$ , a **stationary competitive equilibrium** comprises **functions**

$$\left\{ V(a, y), c(a, y), g(a, y) \right\} \quad \text{and} \quad \left\{ Y, \ell, r, w \right\}$$

such that (i) households optimize, (ii) firms optimize, (iii) markets clear, and (iv) the joint density evolves consistently with household behavior.

- Natural extension of “steady state” concept to HA economies
- Macroeconomic aggregates are constant. Distribution  $g(a, y)$  is constant but households still move around as they draw idiosyncratic income shocks
- Usual notion of “steady” is: “if you start there, you stay there”

# Stationary Competitive Equilibrium Conditions

$$\rho V(a, y_j) = u(c(a, y_j)) + s(a, y_j) \partial_a V(a, y_j) + \lambda_j (V(a, y_{-j}) - V(a, y_j))$$

$$\partial_a V(\underline{a}, y_j) \geq u'(r\underline{a} + Ay_j)$$

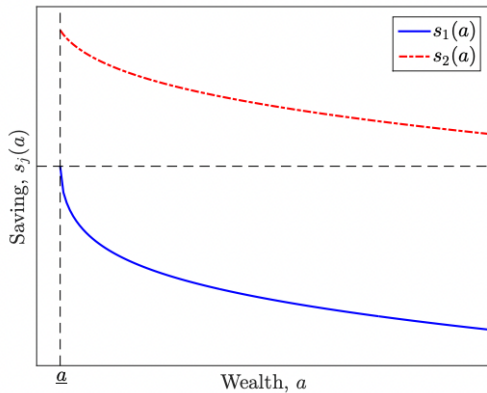
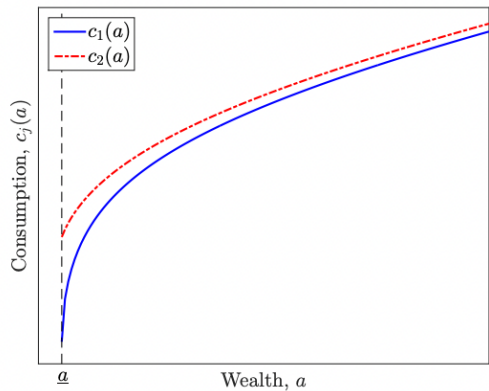
$$u'(c(a, y_j)) = \partial_a V(a, y_j)$$

$$0 = -\partial_a \left[ s(a, y_j) g(a, y_j) \right] - \lambda_j g(a, y_j) + \lambda_{-j} g(a, y_{-j})$$

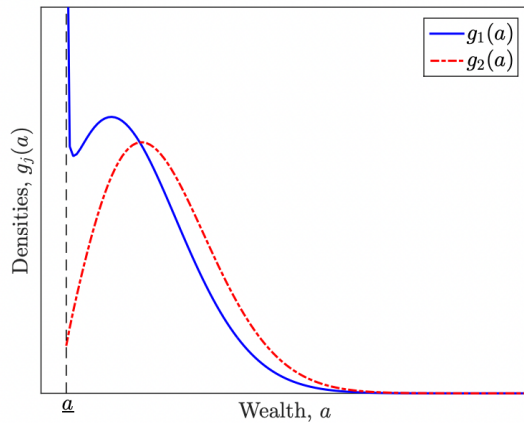
$$0 = \sum_j \int a g(a, y_j) da$$



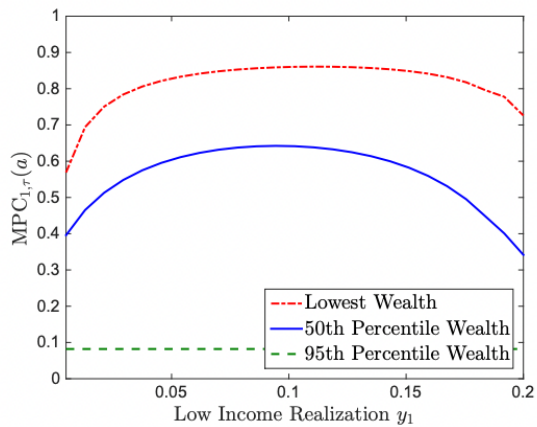
# Typical Consumption and Saving Policy Functions



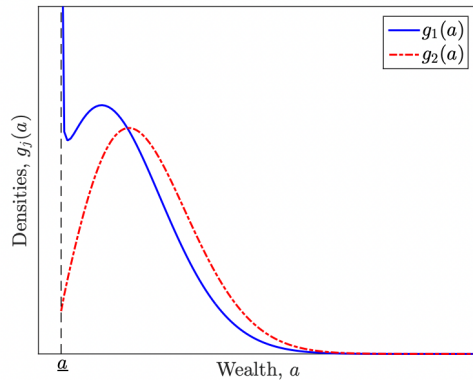
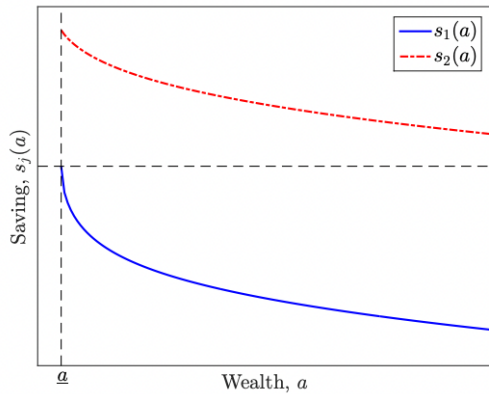
# Typical Stationary Distribution



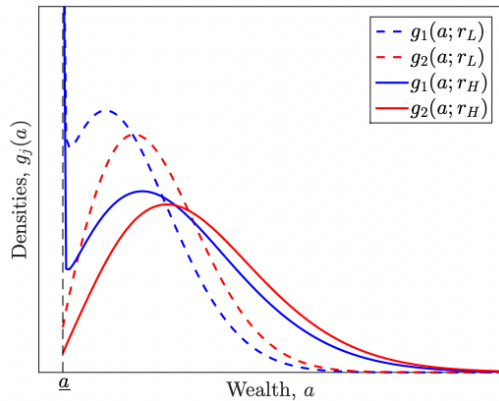
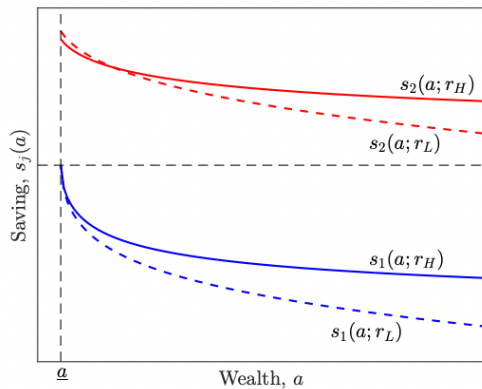
# Household MPC



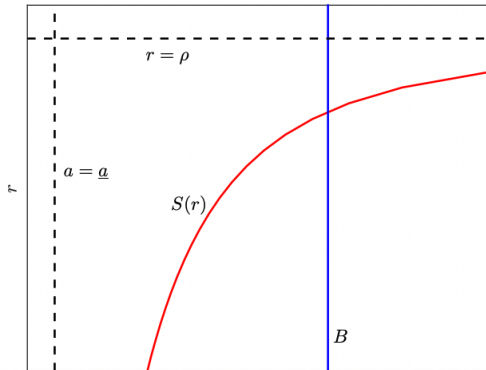
# General Equilibrium: Existence and Uniqueness



Increase in  $r$  from  $r_L$  to  $r_H > r_L$

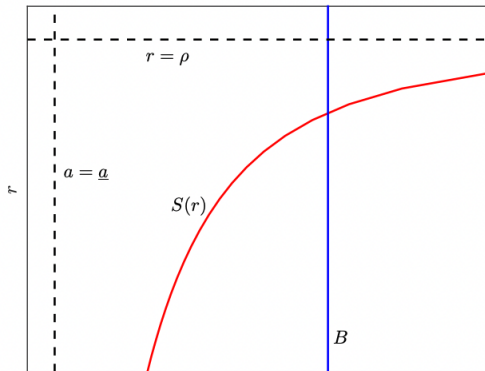


# Stationary Equilibrium



Asset Supply 
$$S(r) = \int_{\underline{a}}^{\infty} a g_1(a; r) da + \int_{\underline{a}}^{\infty} a g_2(a; r) da$$

# Stationary Equilibrium



$$\text{Asset Supply } S(r) = \int_{\underline{a}}^{\infty} a g_1(a; r) da + \int_{\underline{a}}^{\infty} a g_2(a; r) da$$

**Proposition:** a stationary equilibrium exists (also unique!)

## Extension: Diffusion Income Process

